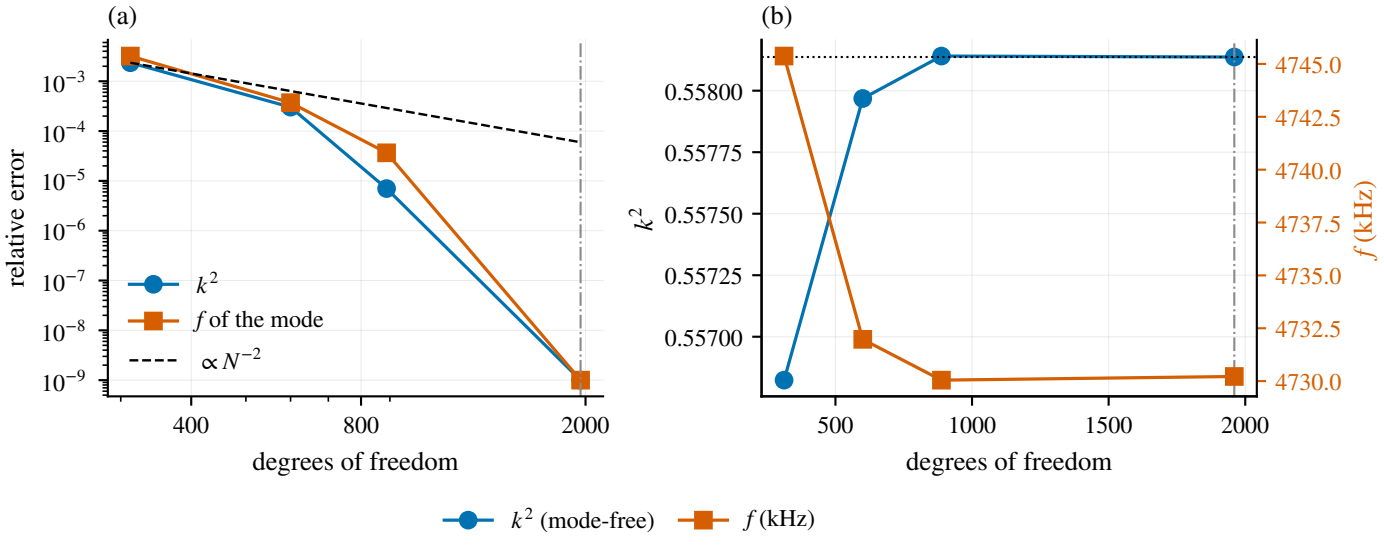
(a)
relative error
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
k^2
f of the mode
$\propto N^{-2}$
400
800
2000
degrees of freedom
(b)
k^2
0.55800
0.55775
0.55750
0.55725
0.55700
f (kHz)
4745.0
4742.5
4740.0
4737.5
4735.0
4732.5
4730.0
500
1000
1500
2000
degrees of freedom
k^2 (mode-free)
f (kHz)